

1. What is static analysis?

- a) Analyzing code during execution
 - b) Analyzing code without executing it ☒
 - c) Testing code with sample inputs
 - d) Debugging runtime errors
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2. Java is considered a:

- a) Dynamically typed language
 - b) Weakly typed language
 - c) Statically typed language ☒
 - d) Untyped language
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3. Static typing allows the compiler to:

- a) Detect errors at runtime
 - b) Deduce types at compile time ☒
 - c) Handle divide-by-zero errors
 - d) Fix logical errors automatically
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4. Which error is typically caught by static checking?

- a) Divide by zero
 - b) Array index out of bounds
 - c) Wrong number of method arguments ☒
 - d) Infinite loop at runtime
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5. Which language uses dynamic typing?

- a) Java
 - b) C++
 - c) Python ☒
 - d) C
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6. Static checking mainly focuses on errors related to:

- a) Specific values
- b) Program logic

- c) Types and syntax ☒
 - d) User input
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7. Which of the following is an example of dynamic checking?

- a) Syntax error
 - b) Wrong method name
 - c) Divide by zero ☒
 - d) Type mismatch
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8. Which of the following can static analysis detect?

- a) SQL injection vulnerabilities ☒
 - b) Logic errors with specific inputs
 - c) Correct output values
 - d) Performance at runtime
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9. Integer division in Java (5 / 2) results in:

- a) 2.5
 - b) 3
 - c) 2 ☒
 - d) Compilation error
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10. Integer overflow in Java:

- a) Causes compilation error
 - b) Causes runtime exception
 - c) Wraps around silently ☒
 - d) Stops program execution
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11. NaN is associated with which data type?

- a) int
- b) long
- c) boolean
- d) double ☒

12. Arrays in Java are:

- a) Variable-length
- b) Fixed-length ☒
- c) Immutable
- d) Dynamically resized automatically

13. Which operation is NOT valid on arrays?

- a) Indexing
- b) Assignment
- c) Changing length ☒
- d) Getting length

14. A potential bug in using a fixed-size array is:

- a) Type mismatch
- b) Overflowing the array length ☒
- c) Syntax error
- d) Compilation failure

15. Which collection is variable-length?

- a) `int[]`
- b) `double[]`
- c) `Array`
- d) `List` ☒

16. Why do we use `List<Integer>` instead of `List<int>`?

- a) Performance reasons
 - b) Lists only support object types ☒
 - c) `int` is deprecated
 - d) `Integer` is faster
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17. Which of the following is an interface?

- a) ArrayList
 - b) List ☒
 - c) Integer
 - d) String
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18. list.size() returns:

- a) Last index
 - b) Maximum capacity
 - c) Current number of elements ☒
 - d) Fixed size
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19. A mutable object is one that:

- a) Cannot be changed
 - b) Is final
 - c) Can be changed after creation ☒
 - d) Is primitive
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20. Which of the following is immutable?

- a) ArrayList
 - b) StringBuilder
 - c) Array
 - d) String ☒
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21. What does mutating a value mean?

- a) Reassigning a variable
 - b) Changing object reference
 - c) Changing contents of a mutable object ☒
 - d) Creating a new object
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22. What does the keyword final mean for a variable?

- a) Its value must be constant
- b) It cannot be reassigned ☒

- c) It must be immutable
 - d) It cannot reference objects
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23. Aliasing occurs when:

- a) Two variables have same name
 - b) Two variables point to same object ☒
 - c) A variable is final
 - d) A method is static
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24. Why are mutable objects risky?

- a) They reduce performance
 - b) They increase memory usage
 - c) Aliasing can cause unexpected changes ☒
 - d) They cannot be debugged
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25. Passing a mutable object to a method may:

- a) Create a copy automatically
 - b) Prevent modification
 - c) Allow unintended modification ☒
 - d) Cause compile-time error
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26. What happens if you structurally modify a list while iterating using for-each?

- a) No issue
 - b) Infinite loop
 - c) ConcurrentModificationException ☒
 - d) Static error
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27. Which tool focuses on Java coding style?

- a) FindBugs
- b) Checkstyle ☒
- c) JUnit
- d) Eclipse

28. FindBugs analyzes:

- a) Source code only
- b) Bytecode ☒
- c) Runtime logs
- d) User input

29. Which FindBugs category relates to SQL injection?

- a) Performance
- b) Dodgy
- c) Security ☒
- d) Correctness

30. Declaring assumptions in code improves:

- a) Compiler speed
- b) Runtime performance
- c) Code readability and safety ☒
- d) Memory usage